

COM3110/4115: Text Processing

Information Retrieval: Task definition & Document Indexing

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- **Definition of the information retrieval problem**
- **Approaches to document indexing**
 - ◇ **manual approaches**
 - ◇ **automatic approaches**
- Automated retrieval models
 - ◇ boolean model
 - ◇ ranked retrieval methods (e.g. vector space model)
- Term manipulation:
 - ◇ stemming, stopwords, term weighting
- Evaluation
- Advanced Topics in IR

- By the end of the IR sessions, you should be able to:
 - ◊ implement a simple IR system (from indexing to retrieval) and
 - ◊ evaluate how well it does on a gold-standard dataset
- First assignment addresses a substantial part of the above

jaguar

[Jaguar International - Market selector page](#)

www.jaguar.com/

Official worldwide web site of **Jaguar** Cars. Directs users to pages tailored to country-specific markets and model-specific websites.

[Jaguar International - Home](#)



www.jaguar.com/gl/en/

8 Jul 2009

Our mission at **Jaguar** has been to create and build beautiful fast cars. The XK, XF, and XJ bring the ...

[More videos for jaguar »](#)

[Jaguar Cars - Wikipedia, the free encyclopedia](#)

en.wikipedia.org/wiki/Jaguar_Cars

Jaguar Cars Ltd, known simply as **Jaguar** is a British luxury and sports car manufacturer, headquartered in Whitley, Coventry, England. It is part of the **Jaguar** ...

[Jaguar - Wikipedia, the free encyclopedia](#)

en.wikipedia.org/wiki/Jaguar

The **jaguar** is a big cat, a feline in the Panthera genus, and is the only Panthera species found in the Americas. The **jaguar** is the third-largest feline after the tiger ...

Google search (contd)

jaguar south america

[Jaguar - Wikipedia, the free encyclopedia](#)

en.wikipedia.org/wiki/Jaguar

The **jaguar's** present range extends from Southern United States and Mexico across much of Central **America** and **south** to Paraguay and northern Argentina.

[Jaguar Cars](#) - [Jaguar \(disambiguation\)](#) - [Jacksonville Jaguars](#) - [Jaguarundi](#)

[Images for jaguar south america](#) - Report images



[South America - Jaguar](#)

library.thinkquest.org/5053/SouthAmerica/jaguar.html

Jaguars are magnificent cats that prowl the **South American** jungles. They are fascinating to learn about! To jump to a section, use our Quick Jump below by ...

[Jaguars, Jaguar Pictures, Jaguar Facts - National Geographic](#)

animals.nationalgeographic.com/animals/mammals/jaguar/

Jaguars are the largest of **South America's** big cats. They once roamed from the southern tip of that continent north to the region surrounding the U.S.-Mexico ...

Google search (contd)

black fast jaguar

[Jaguars, Jaguar Pictures, Jaguar Facts - National Geographic](#)

animals.nationalgeographic.com/animals/mammals/jaguar/

Learn all you wanted to know about **jaguars** with pictures, videos, photos, facts, ... **Fast** Facts. Type: Mammal; Diet: Carnivore; Average life span in the wild: 12 to ... Most **jaguars** are tan or orange with distinctive **black** spots, dubbed "rosettes" ...

[Jaguar XKR black fast on trackday - YouTube](#)



www.youtube.com/watch?v...

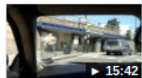
2 Jul 2012 - 16 sec - Uploaded by PrestigeCarCompany

Enjoy this **Jaguar** XKR trackday video.

<http://www.prestigecarcompany.co.uk/> Prestige Super Car Sales

...

[Jaguar XKR Black Pack fast racing - YouTube](#)



www.youtube.com/watch?v...k

16 Aug 2011 - 16 min - Uploaded by MrBobkumar

My XKR being driven hard thru town, hear the sounds of this beast....left window open so noise from air etc....but ...

[Jaguar XKR black fast on trackday. Rear Shot - YouTube](#)



www.youtube.com/watch?v=RmvW...

2 Jul 2012 - 17 sec - Uploaded by PrestigeCarCompany

Enjoy this **Jaguar** XKR trackday video.

<http://www.prestigecarcompany.co.uk/> Prestige Super Car Sales

...

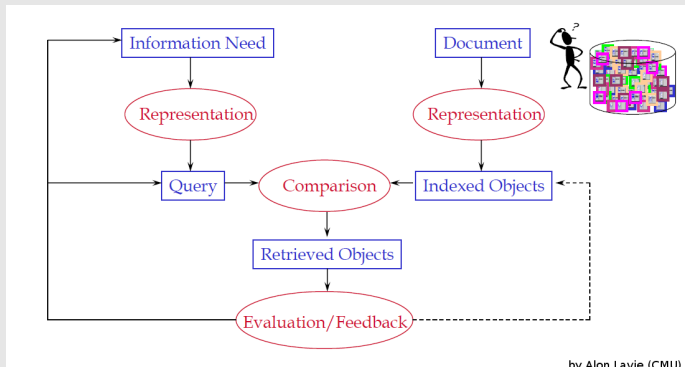
[More videos for black fast jaguar »](#)

Google search (contd)

- What is Google's IR system doing?
 - ◊ Finding pages that contain the words in the query.
- How does it rank these pages?
 - ◊ By “relevance” to the query.
 - ◊ Note: Google may also use information about you, the searcher, depending on various factors. This is not transparent and will be ignored by us.
- How does it do it so fast?
 - ◊ By clever indexing (and a lot of hardware!)
 - ◊ Note: estimated minimum number of indexed pages on the Web as of 27.09.23: **6.43 billion** (<https://www.worldwidewebsize.com>)
Was **3.71 billion** at this time last year and **1.98 billion** the year before.

Information Retrieval: The Task

Text Retrieval: find documents that are “relevant” to a user query.



- **Given:** a large, static document collection
- **Given:** an information need (keyword-based query)
- **Task:** find all and only documents relevant to query

Information Retrieval: The Task

Typical IR systems:

- Search a set of abstracts
- Search newspaper articles
- Search a library or archive collection
- Search a corporate intranet
- Search the Web

Typically: more about statistics than 'language', but the object to retrieve (and process) is language

- How can I formulate a query?
 - ◊ query type: normally keywords, could be a natural language question
- How are the documents represented?
 - ◊ document representation/indexing
- How does the system find the best-matching document?
 - ◊ retrieval model
- How does the system find it *efficiently*?
 - ◊ indexing
- How are the results presented to me?
 - ◊ unsorted list, ranked list, clusters
- How do we know whether the system is any good?
 - ◊ evaluation

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Document Representation/Indexing

The task of choosing **index terms** that describe documents well and organising them to allow documents to be found efficiently

- Manual:

- ◊ index terms assigned by humans using fixed vocabularies
- ◊ labour and training intensive

- Automatic:

- ◊ in theory, all distinct word forms in source text may become index terms
 - But index terms must **only** derive from the source text
- ◊ in practice, various operations performed to improve retrieval accuracy and efficiency
 - Term manipulation (certain words count as the same term)
 - Term weighting (certain terms are more important than others)

- Historically, before widespread digitisation of text, information retrieval was the domain of librarians
 - ◇ Books were assigned to categories in a hierarchical classification scheme
 - ◇ Articles were assigned keywords from subject-specific controlled vocabularies
 - ◇ Typically these schemes/vocabularies were large – thousands of items
- Examples
 - ◇ Dewey Decimal System
 - ◇ Library of Congress Subject Headings
 - ◇ ACM – subfields of CS
 - ◇ MeSH – Medical Subject Headings

Example: Manual Indexing — ACM

ACM Computing Classification System (1998)	
B	Hardware
B.3	Memory structures
B.3.0	General
B.3.1	Semiconductor Memories (NEW) (was B.7.1) Dynamic memory (DRAM) (NEW) Read-only memory (ROM) (NEW) Static memory (SRAM) (NEW)
B.3.2	Design Styles (was D.4.2) Associative memories Cache memories Interleaved memories Mass storage (e.g., magnetic, optical, RAID) Primary memory Sequential-access memory

Example: Manual Indexing — MeSH

MeSH — Medical Subject Headings

- a very large *controlled vocabulary* for describing/indexing medical documents, e.g. journal papers and books
- provides a *hierarchy* of **descriptors** (a.k.a. *subject headings*)
 - ◊ assigned to documents to describe their content
- hierarchy has a number of *top-level* categories, e.g.:
 - ◊ Anatomy [A]
 - ◊ Organisms [B]
 - ◊ Diseases [C]
 - ◊ Chemicals and Drugs [D]
 - ◊ Analytical, Diagnostic and Therapeutic Techniques and Equipment [E]
 - ◊ Psychiatry and Psychology [F]
 - ◊ Biological Sciences [G]

...

Example: Manual Indexing — MeSH (contd)

- And a number of subcategories (more specific/detailed terms):
 - Diseases [C]
 - MeSH C01 --- bacterial infections and mycoses
 - MeSH C02 --- virus diseases
 - MeSH C03 --- parasitic diseases
 - MeSH C04 --- neoplasms
 - MeSH C05 --- musculoskeletal diseases
 - MeSH C06 --- digestive system diseases
 - MeSH C07 --- stomatognathic diseases
 - MeSH C08 --- respiratory tract diseases
 - MeSH C09 --- otorhinolaryngologic diseases
 - MeSH C10 --- nervous system diseases
 - MeSH C11 --- eye diseases
 - MeSH C12 --- urologic and male genital diseases
 - MeSH C13 --- female genital diseases and pregnancy complications
 - MeSH C14 --- cardiovascular diseases

Example: Manual Indexing — MeSH (contd)

- And a number of subsubcategories (even more specific/detailed terms):

Eye Diseases [C11]

Asthenopia [C11.093]

▶ Conjunctival Diseases [C11.187]

Conjunctival Neoplasms [C11.187.169]

Conjunctivitis [C11.187.183] +

Pterygium [C11.187.781]

Xerophthalmia [C11.187.810]

Corneal Diseases [C11.204] +

Eye Abnormalities [C11.250] +

Eye Diseases, Hereditary [C11.270] +

Eye Hemorrhage [C11.290] +

Eye Infections [C11.294] +

Example: Manual Indexing — MeSH (contd)

- And a number of subsubsubcategories (yet again more specific/detailed terms):

Eye Diseases [C11]

Conjunctival Diseases [C11.187]

Conjunctival Neoplasms [C11.187.169]

► Conjunctivitis [C11.187.183]

Conjunctivitis, Allergic [C11.187.183.200]

Conjunctivitis, Bacterial [C11.187.183.220] +

Conjunctivitis, Viral [C11.187.183.240] +

Keratoconjunctivitis [C11.187.183.394] +

Reiter Syndrome [C11.187.183.749]

Pterygium [C11.187.781]

Xerophthalmia [C11.187.810]

Example: Manual Indexing — MeSH (contd)

- MEDLINE — Medical Literature Analysis and Retrieval System Online
 - ◇ international database of literature for medicine and the life sciences
 - ◇ includes papers from around 5200 different sources (mostly journals), in around 40 languages
 - ◇ database now holds records for > 29 million papers
 - ◇ See https://www.nlm.nih.gov/medline/medline_overview.html
- Each MEDLINE article indexed with 10-15 descriptors from MeSH
 - ◇ papers accessed by PubMed search engine interface, using MeSH terms (and other terms, e.g. author name, etc)
 - ◇ by default, all descriptors below a given one in the hierarchy are also included in search

- Advantages:

- ◊ High precision searches
- ◊ Works well for closed collections (books in a library)

- Problems:

- ◊ Searchers need to know terms to achieve high precision
- ◊ Labellers need to be trained to achieve consistency
 - Not feasible to expect this from all content creators on the web
- ◊ Collections are dynamic → schemes change constantly
- ◊ Expensive

- No predefined set of *index terms* that must be manually assigned
- Instead: use **natural language** as indexing language
- Words in the document give information about its content
- Implementation of indices: **inverted files**
- This is what modern search engines do
 - ◊ at least, it's an important **part** of the story

- A small collection of documents ...

<i>Document</i>	<i>Text</i>
1	Pease porridge hot , pease porridge cold
2	Pease porridge in the pot
3	Nine days old
4	Some like it hot , some like it cold
5	Some like it in the pot
6	Nine days old

Say we want to search for word **hot**. How do we do it?

Inverted files

- A basic **inverted file index**

- ◇ records for each term, the ids of the documents in which it appears
- ◇ only matters if it **does** or **does not** appear – not how many times

<i>Doc</i>	<i>Text</i>
1	Pease porridge hot, pease porridge cold
2	Pease porridge in the pot
3	Nine days old
4	Some like it hot, some like it cold
5	Some like it in the pot
6	Nine days old



<i>Num</i>	<i>Token</i>	<i>Docs</i>
1	cold	1, 4
2	days	3, 6
3	hot	1, 4
4	in	2, 5
5	it	4, 5
6	like	4, 5
7	nine	3, 6
8	old	3, 6
9	pease	1, 2
10	porridge	1, 2
11	pot	2, 5
12	some	4, 5
13	the	2, 5

Inverted files (contd)

- A more sophisticated version ...
 - ◇ also record **count** of occurrences within each document
 - ◇ help find documents **more relevant** to query

<i>Doc</i>	<i>Text</i>
1	Pease porridge hot, pease porridge cold
2	Pease porridge in the pot
3	Nine days old
4	Some like it hot, some like it cold
5	Some like it in the pot
6	Nine days old



<i>Num</i>	<i>Token</i>	<i>Docs</i>
1	cold	1:1, 4:1
2	days	3:1, 6:1
3	hot	1:1, 4:1
4	in	2:1, 5:1
5	it	4:2, 5:1
6	like	4:2, 5:1
7	nine	3:1, 6:1
8	old	3:1, 6:1
9	pease	1:2, 2:1
10	porridge	1:2, 2:1
11	pot	2:1, 5:1
12	some	4:2, 5:1
13	the	2:1, 5:1

Inverted files (contd)

- An even more sophisticated version ...
 - ◇ also record **position** of each term occurrence within documents
 - ◇ may be useful for searching for **phrases** in documents

<i>Doc</i>	<i>Text</i>
1	Pease porridge hot, pease porridge cold
2	Pease porridge in the pot
3	Nine days old
4	Some like it hot, some like it cold
5	Some like it in the pot
6	Nine days old



<i>Num</i>	<i>Token</i>	<i>Docs</i>
1	cold	1:(6), 4:(8)
2	days	3:(2), 6:(2)
3	hot	1:(3), 4:(4)
4	in	2:(3), 5:(4)
5	it	4:(3, 7), 5:(3)
6	like	4:(2, 6), 5:(2)
7	nine	3:(1), 6:(1)
8	old	3:(3), 6:(3)
9	pease	1:(1, 4), 2:(1)
10	porridge	1:(2, 5), 2:(2)
11	pot	2:(5), 5:(6)
12	some	4:(1, 5), 5:(1)
13	the	2:(4), 5:(5)

- **Information Retrieval (IR)** is the task of retrieving those natural language (NL) documents from a large collection of such documents that are **relevant** to a user query, typically expressed as a small number of NL keywords
- IR is a key technology in the digital age, allowing effective and efficient access to information stored in massive collections of documents on the web, in archives and in corporate intranets.
- A key process underlying IR systems is **indexing**
 - ◇ indexing may be **manual** or **automatic**
 - ◇ automatic indexing is the only practicable solution in most settings

Summary (cont)

- Automatic indexing involves
 - ◊ selecting terms from a document to effectively represent the document's content
 - ◊ structuring those terms to allow efficient retrieval
- The most common index structure for IR is the **inverted index**
- An inverted index associates terms with the documents in which they occur and optionally also records their frequency or offset with the document

References

- Jurafsky, Daniel and James H. Martin. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition.
 - ◇ 2nd ed. is available as an e-book from the University Library.
 - ◇ 3rd ed. in draft form (much more up to date) is available at: <https://web.stanford.edu/~jurafsky/slp3/>. See especially, in the 3rd ed., Chapter 14 and Chapters 5-12 for background on neural methods in NLP.
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